

Pryor on *De Jure* Codesignation

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1. A paradox about attitude reports

Mary is someone who will confidently tell you ‘Cicero is wise, whereas Tully is foolish’. Let ‘F’ abbreviate some predicate uniquely singling out Cicero (=Tully), e.g. ‘author of *De Finibus*’, and let A–D abbreviate the following sentences:

- A Every F is such that Mary believes he is wise, and every F is such that Mary believes he is foolish.
- B Every F is such that Mary believes he is wise and believes he is foolish.
- C Every F is such that Mary believes he is wise and he is foolish.
- D Every F is such that Mary believes he is wise and foolish.

Our paradox then consists of the following five jointly inconsistent, but individually plausible sentences:

A	If A then B	If B then C	If C then D	Not D
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2. The Putnam-Richard-Fine solutions

Jim’s central topic is a view whose hallmark is a willingness to give up on $A \rightarrow B$ and/or $B \rightarrow C$ (at least on some important readings).

- The falsity of C is explained by appeal to [F2] and [F3]:
 - [F2] Multiple variable occurrences bound by a single quantifier are always coordinated.
 - [F3] On some (readings of) attitude reports, expressions in their complement are coordinated iff they attribute cyclic thoughts.
- The falsity of $A \rightarrow B$ (on some readings) stems from an independent component of Fine’s view, also analogous to something Richard says:
 - [F4] Fine allows coordination across the complements of multiple attitude verbs.

Others who are willing to give up $B \rightarrow C$ are David Kaplan (in his super-influential paper ‘Quantifying In’, not just the more obscure pieces Jim mentions), and several recent theorists who have attempted to give compositional implementations of Kaplan-inspired

truth conditions for attitude reports: Percus and Sauerland¹, Ninan², Charlow and Sharvit³, Rieppel⁴. Meanwhile, Yalcin⁵ explores a view that on which $B \rightarrow C$ holds but $A \rightarrow B$ fails. Lederman⁶ develops another such view, modelled on a proposal of Maria Aloni⁷.

- All these authors differ from Fine and Richard in holding that the phenomenon of ‘rational inconsistent belief’ illustrated by sentence A arises *only* for belief-reports containing *externally bound variables* (like the occurrences of ‘he’ in A). They would thus not give a parallel response to parallel-looking paradoxes involving names, such as the paradox consisting of B', $B' \rightarrow C'$, $C' \rightarrow D'$, and $\neg D'$, where B', C', D' are as follows:
 B' Mary believes Cicero is wise and Mary believes Cicero is foolish.
 C' Mary believes Cicero is wise and Cicero is foolish.
 D' Mary believes Cicero is wise and foolish.
- Well *actually* some of these authors, e.g. Rieppel, *do* think that names behave in a parallel way to (locally free) variables; they respond by saying that *names are variables*.⁸ Like Fine and Richard, I think that A-type behavior due to ‘identity confusion’ can in principle arise with just about any word (‘water’, ‘lawyer’, ‘massive’, ‘melt’) so I think that special pleading about names isn’t going to help.

3. Other solutions

It’s widely held that belief-attributions like ‘Every F is such that Mary believes he was eloquent’ are context-sensitive—true on some readings, false on others.⁹ Anyone who accepts this view should think that A is false on *some* interpretations: the really interesting question is which of $A \rightarrow B$, $B \rightarrow C$, $C \rightarrow D$, and $\neg D$ is false *on the interpretations that make A true*.

¹ Orin Percus and Uli Sauerland, ‘On the LFs of Attitude Reports’. Their idea (for which they credit lecture notes by Heim) is that every occurrence of an attitude verb introduces a separate existential quantification over “concept-generators”—functions from objects to individual concepts under which they fall. Like the other authors mentioned they are thinking of these in a descriptivist way, but one could implement the thought using other ways of cashing out ‘guise’ talk, including ones in terms of mentalese sentences.

² Dilip Ninan, ‘Counterfactual attitudes and multi-centered worlds’.

³ Simon Charlow and Yael Sharvit, ‘Bound de re pronouns and the LFs of attitude reports’.

⁴ Michael Rieppel, ‘Names, Masks, and Double Vision’.

⁵ Seth Yalcin, ‘Quantifying In from a Fregean Perspective’.

⁶ Harvey Lederman, ‘Reply to Yalcin’.

⁷ Maria Aloni, ‘Individual concepts in modal predicate logic’.

⁸ This view is originally defended in Sam Cumming, ‘Variabilism’.

⁹ For arguments see Stephen Schiffer, ‘Naming and Knowing’ and Dorr, ‘Transparency and the Context-Sensitivity of Attitude Reports’.

- Some theorists don't think that A is true on *any* uniform interpretation—making it true requires a “mid-sentence context change”. Examples: Robert Stalnaker, Maria Aloni. Other theorists suggest that A is unambiguously false (when quantifiers are unrestricted), although its existentially quantified counterpart ‘Some F is such that Mary believes he is wise and some F is such that Mary believes he is foolish’ is unambiguously true.¹⁰
- Some theorists¹¹ have recently suggested a view on which $A \rightarrow B$, $B \rightarrow C$, and $C \rightarrow D$ are true on all (uniform) interpretations and A is true on some, so that D is true on some interpretations.
- Bach and Lasnik and Salmon and Soames (see section 1 of Jim's paper) think that $C \rightarrow D$ is just false while A, $A \rightarrow B$, $B \rightarrow C$ and $\neg D$ are just true. A variant of this view (that I like better) admits context-sensitivity, saying that there are *also* interpretations where $A \rightarrow B$, $B \rightarrow C$, $C \rightarrow D$ and $\neg D$ are all true.

4. The Salmon-Soames view

Those who reject $C \rightarrow D$ deny that C has a reading where it (a) requires Mary to go in for what Jim dubs “cyclic thought”, the kind of thing which would tend to go along (in a rational person) with believing that someone is eloquent and wise, but (b) is still permissive with regard to the particular “guise” involved in this cyclic thought, e.g. would be true both in a situation where Mary would say ‘Tully was eloquent and Tully was wise’ and in one where she would say ‘Cicero was eloquent and Cicero was wise’.

- $C \rightarrow D$ deniers may agree that C *tends to suggest* that Mary has a cyclic belief (even when interpreted guise-permissively). But they'll treat this as merely a pragmatic suggestion.

You might suppose that this would require, by parity of reasoning, denying that (1) and (2) have cyclic-thought-requiring readings.

- (1) Every F is such that Mary believes he admired himself
- (2) Mary believes Cicero admired himself

But no! Bach and Lasnik would (I assume) say this. But not Salmon and Soames. According to them, (1) and (2) admit logical forms like (1') and (2'):

- (1') $[\text{Every } F]_1 [\text{Mary believes } [\text{he}_1 [\lambda x_2. x_2 \text{ admired } x_2]]]$
- (2') $\text{Mary believes } [\text{Cicero } [\lambda x_2. x_2 \text{ admired } x_2]]]$

and when disambiguated in this way, they are just false in the scenario where Mary would say ‘Cicero and Tully admired one another but neither admired himself’.

¹⁰ Michael Caie, Jeremy Goodman, and Harvey Lederman, ‘Classical Opacity’ (unpublished MS)

¹¹ Jeremy Goodman and Harvey Lederman, ‘Verbalism’ (unpublished MS).

- Why are (1') and (2') false in this scenario? Because in them, the complement of 'believes' contains only *one* name or locally free variable. According to the truth conditions favoured by Salmon and Soames, each occurrence of a name or locally free variable under an attitude verb introduces an independent existential quantifier.
- Why don't they think that C, similarly, could have an LF like this, and thus require cyclic belief after all:

C' [Every F]₁ [Mary believes [he₁ [$\lambda x_2.x_2$ was eloquent and he₂ was wise]]

Answer: this logical form is ruled out by general considerations about when 'DP raising' is allowed. 'No orator is wise and he is eloquent' can't mean '[No orator]₁ [x_1 is wise and x_1 is eloquent]'.

5. Pryor's objection to S&S

Jim's section 1 challenges S&S by presenting cases which seem *intuitively on a par with* (1) and (2), but where their strategy for dealing with (1) and (2) does not apply: there is no principled way of getting the pronouns in the complement of 'believes' to be *locally bound*. Jim thinks that each of these sentences has a reading that involves *de jure codesignation*, so that embedding the whole thing under 'Mary believes that' would require Mary to have a cyclic thought.

[3d] Terentia betrayed Cicero, so he was unhappy.

[4a] Cicero mourned his daughter, but Atticus didn't [i.e. didn't mourn Cicero's daughter]

[4b] Cicero admired himself, but Atticus didn't [i.e. didn't admire Cicero]

[11a] Only Cicero mourned his daughter [i.e. no-one else mourned Cicero's daughter]

[12] Terentia didn't admire Cicero, but he did

For [3d] the obstacle to a "local binding" LF is the same as for C. In [4a], [4b], and [12], the obstacle comes from a standard theory of ellipsis (endorsed by S&S, cf. Jim's "Z1"), according to which if we want the first conjunct of [4a] to have the LF 'Cicero [$\lambda x_1.x_1$ mourned his₁ daughter]', 'Atticus didn't' would have to be equivalent to Atticus [$\lambda x_1.x_1$ didn't mourn his₁ daughter]'. For [11a], the obstacle comes from a theory of 'only'. I'm not so sure that it would be so bad for S&S to posit the fancy structure they would need to get around this.)

6. Pryor's defence of Putnam-Richard-Fine

Two main objections to the Putnam-Richard-Fine stance come up in Jim's paper.

The first is a general feeling that there is something 'alien' and 'magical' about the kind of sensitivity to patterns of co-ordination the stance requires. Jim's section 7 is designed

to address this by getting us used to formal languages where such behaviour is wired in. Works for me!

The second is a concern emphasised by Soames, namely that there are cases where we have multiple occurrences of a pronoun under an attitude (or speech) verb, bound by the same quantifier, but cyclic thought is not required

[27b] Each man told Maria that he wasn't that man

(3) Each man told Maria that he hated everything he loved

Fine suggests dealing with such examples by invoking a distinct 'de re' reading of the attitude verb that doesn't care about co-ordination inside its argument. Soames objects that this gives us a blanket yes-or-no choice, whereas we should want (if we're treating the phenomenon semantically at all!) to be able to treat some but not other variables bound by the same quantifier as cyclic-thought-requiring.

(4) Each man told Maria that someone he hated had shown him something he loved.

An alternative to appealing to a special reading of the verb would be to give up F2, allowing different pronoun-occurrences bound by the same quantifier sometimes not to count as co-ordinated with one another for purposes of applying F3. But this raises problems when we consider iterated reports like the following (see Jim's note 57)

(5) The children thought that Juan told Maria that he wasn't him.

We want to somehow say that 'he' and 'him' are co-ordinated for the purposes of 'thought' but not for the purposes of 'told'.

Jim (note 57) suggests what a framework for representing these kinds of structural ambiguities might look like (it involves treating some λ s in the ordinary way and others as analogous to $\text{alias}\lambda$ in Scheme, or alternatively introducing a value operator in certain positions). I'm willing to believe it can be done, although right now I don't see how to do flexibly enough to, e.g., get four readings for (6).¹²

(6) Every man told Maria that Juan wanted Mercedes to think that everyone who loved him hated him.

7. A different reason to worry about giving up $A \rightarrow B$

My own impulse to accept $A \rightarrow B$ comes from logical conservatism. Denying the validity of $A \rightarrow B$ requires denying that *universal quantification commutes with conjunction*—i.e. rejecting the validity of the following schema

¹² The readings could be labeled as CCC, CCU, CUU, UUU, where 'C' is 'co-ordinated' and 'U' is unco-ordinated. For example, CCU would be a reading on which every man is saying to Maria 'Juan wants me to be a person x concerning whom Mercedes is subject to identity confusion and thus thinks (non-cyclically of course) that everyone who loves x hates x '.

Every F is such that ϕ ; every F is such that ψ ; so every F is such that ϕ and ψ
In fact view requires giving up a huge chunk of standard quantifier logic, including schemas like these:

Some F is a G; every G is H; so some F is H

(let G = 'author of *De Finibus*', F = 'person believed by Mary to be wise', H = 'believed by Mary to be foolish').

- The denier of $A \rightarrow B$ will reply that the temptation to regard these as unrestrictedly valid is just a mistaken overgeneralisation; the real generalisation in the vicinity has to carve out an exception for the case where the conjunction introduces new coordination.
- Do I have any actual *argument* here other than flat-footed conservatism? Well, I do think there is a good methodological case for giving substantial weight to people's dispositions to endorse or reject sentences based just on their "logical structure", i.e. just noticing a certain pattern of repetitions rather than considering how to interpret every element.

8. Keep $A \rightarrow B$ and just give up $B \rightarrow C$?

There's a similar argument for $B \rightarrow C$ based on the following schema:

Belief Agglomeration: If x believes that ϕ and x believes that ψ then x believes that (ϕ and ψ)

However there are powerful independent arguments against the validity of Belief Agglomeration. These arguments do not threaten the validity of the following schema, which would be enough to run the argument:

Confidence Agglomeration: If x is very confident that ϕ and x is very confident that ψ and x is sufficiently rational then x is moderately confident that (ϕ and ψ)

But maybe it wouldn't be so bad to give up even this weaker schema.

I do however have the sense that there's something arbitrary about a view that builds co-ordination into the semantics in the way required to get C to be false, but doesn't do the same for the likes of B. The mechanism that makes us draw inferences about "internally coordinated thought" from B-type sentences seems very similar to the mechanism that makes us draw such inferences from C-type sentences: it seems unprincipled to build it into the semantics for C-type sentences while falling back on general-purpose pragmatics for B-type sentences.